

1.4.2 Predicates

Statements involving variables, such as " $x > 3$," " $x = y + 3$ " " $x + x = z$," and "Computer x is under attack by an intruder." and "Computer x is functioning properly," are often found in mathematical assertions, in computer programs, and in system specifications. These statements are neither true nor false when the values of the variables are not specified.

The statement " x is greater than 3" has two parts. The first part, the variable x , is the subject of the statement. The second part — the predicate, " $\text{is greater than } 3$ " — refers to a property that the subject of the statement can have. We can denote the statement " x is greater than 3" by $P(x)$, where P denotes the predicate " $\text{is greater than } 3$ " and x is the variable. The statement $P(x)$ is also said to be the value of the propositional function P at x . Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value.

Example 1: Let $P(x)$ denote the statement " $x > 3$ ". What are the truth values of $P(4)$ and $P(2)$?

Soln: $P(4): 4 > 3 \rightarrow \text{True}$ $P(2): 2 > 3 \rightarrow \text{False}$

Example 2: Let $A(x)$ denote the statement "Computer x is under attack by an intruder." Suppose that of the computers on campus, only CS2 and MATH1 are currently under attack by intruders. What are the truth values of $A(\text{CS1})$, $A(\text{CS2})$ and $A(\text{MATH1})$?

Soln: $A(\text{CS1}): \text{Computer CS1 is under attack by an intruder} \rightarrow \text{False}$
 $A(\text{CS2}): \text{Computer CS2 is under attack by an intruder} \rightarrow \text{True}$
 $A(\text{MATH1}): \text{Computer MATH1 is under attack by an intruder} \rightarrow \text{True}$

Example 3: Let $Q(x, y)$ denote the statement " $x = y + 3$." What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Soln: $Q(1, 2): 1 = 2 + 3 \Rightarrow 1 = 5 \rightarrow \text{False}$
 $Q(3, 0): 3 = 0 + 3 \Rightarrow 3 = 3 \rightarrow \text{True}$